

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000;    % Passband cutoff (Hz)

fs = 4000;    % Sampling frequency (Hz)

N = 50;       % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%% --- Plot Frequency Responses ---

figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');
```

